The Open Source Firefighter

The Training Range — A Vision for the Program's School

v1.0 · August 2026

*Document ID: **OSF-3.5** · one document of the Open Source Firefighter specification family.*

> **A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.**

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This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

What this document is

The rest of this family specifies a system: the box, the mesh, the water, the doctrine. This document answers the question that follows every good box onto the pallet: **how do people at every level learn to use it?** The homeowner's answer is easy — the Overview and the Adoption Briefing carry it. The commercial installer's answer is the default — the trades already read specifications for a living. This document is written for the harder and more valuable audience: **the fire and forestry professional**, the district office, the volunteer department, the burn boss, the range manager — the people who will run the coordinated defense at scale, and who need somewhere to train on it.

It is a vision document. The program it describes does not exist yet. It is published now, with the rest of the family, because the ground it describes already exists, sits idle at the exact moments it is most relevant, and belongs — in the ways that matter — to more people than currently realize they are its stakeholders. If you work in a forestry office or run a volunteer department anywhere near a military installation, this document's purpose is that you finish it knowing something you did not know when you started: **you have an avenue onto that ground, and reasons to use it.**

The campus already exists

The Department of Defense operates the largest inventory of instrumented, maintained, fire-relevant training land in the country. Every terrain type. Every fuel condition. Every size, from a pistol range to maneuver areas the size of counties. Roads, water points, impact-area discipline, and — on modern instrumented ranges — installed thermal camera networks and observation towers, maintained on someone else's budget, watching ground where fires start routinely because starting fires is what tracer, artillery, and demolition training does.

And here is the fact this document turns on: **during red flag conditions and high fire danger, those ranges shut down.** Live fire stops. Some years, even blanks stop. The installations' training capacity goes idle at precisely the moment fire training has maximum urgency and realism — and the crews reduced to dry-fire theater on a silent range are sitting in the middle of the best wildland-fire classroom in the country, on the exact days it can teach.

A range that cannot shoot can host node emplacement drills, wet-line lays, buffered-supply-point exercises, UAS patrol circuits, mock-fire coordination problems, and the full surveillance-and-release curriculum described below — none of which require a single live round or an open burn window. The fire restriction that suspends the range's primary mission creates its availability.

The rent is the fuels backlog

Every range in the system is behind on land management. Grass and fuels work is the orphan task of range operations — the busy-work everyone pretends is not their responsibility, deferred every year because the people and money are never there. Meanwhile the interagency burn crews that could treat it arrive on a schedule set months in advance: a few days per facility per spring, weather permitting. Some years the grass is wet and the visit produces nothing. The good years, the crew burns what is reachable in the time available — the flat infield grass, the course roads — and moves to the next post. The hard ground stays hard forever: the draws, the tracer fans under the gun-target lines, the terrain that actually sets fires all summer.

The arithmetic is the problem, not the people. **The scarce resource is not the burn crew; it is the burnable window** — the scattered handful of days each year when a given problem area is safely treatable — and a traveling crew scheduled months ahead almost never overlaps it. A defense that must be present when the window opens

cannot be built from a scarce expert resource that travels. It has to be standing, local, and ready — which is this specification's entire thesis, pointed at prescribed fire.

So the school pays its rent in the host's own currency: **every training evolution leaves the range with fuels work done.** Wet-line exercises hold real burns on real backlog. Node emplacement drills pre-position real terrain defense. The range's thousand-acre problem is the curriculum, and the range officer stops tolerating the program and starts requesting it.

The gap the school closes

Coordination between the fire world and the range world exists today — but as deconfliction, not integration. The burn crew takes the range for its scheduled days; range control clears the space and steps back; everyone follows the process. The information flows one way. Nobody asks the range office where the fires usually start, which draws always get away, or what instrumentation is buried under which grass — and so target pits and lifters have burned under professionally executed prescribed fire, not because anyone failed, but because the professional-courtesy structure defined the locals as a scheduling entity rather than a knowledge source. Every range manager can draw the ignition map of his own ground from memory. Almost no one from the fire side has ever asked.

Deconfliction keeps two operations from colliding. Integration lets them inform each other. The missing piece is not a coordinator — the coordination slot is filled. The missing piece is *standing*: someone the fire side recognizes as a peer, who can put local ground truth inside the burn plan instead of outside the tape. Credentials are the carrier wave for knowledge; the same warning lands differently from "a local" than from a dual-qualified professional whose input the burn boss must weight.

That is the school's product. Not a facility. Not a technology. **A profession**: operators fluent in both dialects, minted deliberately instead of occurring by accident.

The two formations that never overlapped

The wildland fire tradition produces superb ground-readers: professionals who read fuel, weather, slope, and fire behavior as a first language, formed by seasons on the line. That formation is analog and terrestrial by lineage, and it is honorable. What it never includes — because the equipment does not exist in that world — is the instrumented battlespace: mesh networks, thermal camera grids, observation towers, coordination protocols, and the discipline of running a surveillance picture.

The range world produces the complement: operators who run multi-camera thermal networks as a daily instrument, who detect a developing situation in a wood line before it becomes a problem, and who live inside the loop of *detect, correlate, decide, task*. On instrumented ranges, that discipline extends to its full form — an observer sensor, a decision authority, and an effector as separate nodes on a network, with positive identification established across independent sensors and consequential action released only by a human voice, and held the instant anything is ambiguous. That reflex cannot be read out of a book. It is built by running it, and the only places it can be run are exactly the ranges this document describes.

The curriculum, in one line: hand the ground-reader the grid. Take the fire professional who already reads the land and teach the surveillance picture, the correlation call, the release discipline, and the coordinated water employment this specification defines — talk the student onto the thermal signature, require agreement across independent sensors, require the words said out loud, and teach that holding everything when unsatisfied is not caution but the job. The military spent a century writing this pedagogy. The school ports it to fire — dry, then rehearsed, then live — and the dry tier runs on any instrumented range on any red flag day, when nothing else can train at all.

What standing local capability unlocks

A range operating this specification's terrain layer — pre-surveyed node sites, reversible wet line, buffered supply points fed by ordinary porta-tank orders through local departments and forestry, the installed thermal net as overwatch, a resident cadre of dual-fluent operators — can prosecute its own fuels program opportunistically: hit the hard draw on the exact right Tuesday in November, one problem area at a time, all year, with the traveling professional crews reserved for the genuinely difficult prescriptions where their depth is irreplaceable. The lighting was never the hard part of prescribed fire; the *holding* is — and pre-positioned water collapses holding risk on exactly the ground nobody will otherwise touch: the deep masticated mats, the smoldering-prone cleared units where grass grows through years-old hydroaxe work, the category of ground that needs fire most and receives it least.

The sequence is deliberately modest: **assist first.** The program shows up as the holding and safety element that makes the interagency crew's scheduled days count. Capability migrates the way it always migrates — toward whoever does the work best with the least friction — and no one needs to be displaced for the backlog to start shrinking.

You are a stakeholder in that ground

This is the part written directly to the reader in the forestry office and the volunteer fire station. The access avenues exist now:

- **Any volunteer fire department in the country has a path onto a military installation** — mutual aid relationships, installation support agreements, and joint training arrangements are established machinery, exercised routinely.
- **Federal and cooperating agencies have partnership avenues** to installation ranges as a matter of standing policy; the ground is government ground, and cooperative use is what the machinery is for.
- **The specialized facilities sit empty.** Urban training complexes — dozens of varied structures, furnished, built for search and maneuver training — stand unused the overwhelming majority of the year, while departments an hour away improvise structure-search training in whatever they can borrow.
- **The ranges' fire problem is real and unowned**, which means the agency or department that shows up with capability is not competing for a resource — it is answering a need the host cannot currently meet.

The ask of this document is small: the next time you drive past the installation boundary, do not think of it as someone else's fence. Think of it as the best-instrumented fire classroom in your county, idle on red flag days, behind on the exact work your profession exists to do, reachable through agreements your organization is already entitled to pursue. The coordinated defense this family specifies will be run, at scale, by people — and the people will be trained somewhere. The somewhere already exists. It is waiting for its stakeholders to recognize themselves.

Different colors of the same money

One more thing the reader in the district office already knows, said out loud: the volunteer department, the range, the Army, city fire, EMS, police, forestry, the hotshot crew, BLM, state fire, federal fire, county fire, the HOA, the FAA, the water department, and the power company are **different colors of the same public money.** The citizen writes one check, and it comes back wearing twenty uniforms that each defend a slice as though it were separate funds. It never was. The fragmentation everyone treats as natural law is accounting made visible as turf — and the public that pays for all the sleeves does not care whose patch is on one when the ridge is burning.

This specification's coordination layer is built for exactly that reality, in exactly one role: **the box is the secretary.** Not a commander, not a new agency, not anyone's boss — the secretary that did the interagency coordinating in advance, mechanically, so that every professional simply does their job. The hydrant, the tender, the terrain node, the range's thermal net, and the incident commander's tablet cooperate through a protocol that already reconciled what the org charts never will, without a memorandum summit, without a turf negotiation, without anyone yielding authority to anyone. Coordination as infrastructure instead of meetings. If the public wants the red boxes — and understands that it is all the same game, played by the same team, on the same dollar — then the remaining work is what this document describes: train the team on the shared ground the team already owns.

What this document is not

It requests no facility, no construction, no program office, and no budget line. It proposes tenancy, not ownership — conducted in the tenant ethic this specification's author practiced for eleven years as a range manager: honor the host's load-bearing rules absolutely, leave the ground better than found, and pay rent in solved problems. The program described here begins the way everything in this family begins: with existing parts, existing avenues, and the arithmetic of assembling what is already paid for.

The graduate this school produces does not exist yet. The ground that will produce them does. That asymmetry is an invitation.